ModelIt!

Teacher's Guide

How LeBron Turns Food Into Dunks

The Cellular Science Behind Athletic Performance

Quick Reference

Grade Level	Grade 8
NGSS Standard	MS-LS1-5, MS-LS1-7
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-LS1-5, MS-LS1-7

Construct a scientific explanation based on evidence for how environmental and genetic factors influence the growth of organisms. Develop a model to describe how food is rearranged through chemical reactions forming new molecules that support growth and/or release energy as this matter moves through an organism.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model to describe how food molecules are rearranged through chemical reactions to release energy.
Disciplinary Core Idea	**LS1.C Organization for Matter and Energy Flow in Organisms** — Within individual organisms, food is broken down through chemical reactions that rearrange molecules and release energy.
Crosscutting Concept	**Energy and Matter** — Students trace matter and energy as food molecules are transformed through cellular respiration into ATP energy and waste products.

CAST Testing Alignment

MULTIPLE CHOICE:

During cellular respiration, glucose and oxygen are converted into carbon dioxide, water, and ATP. What type of process is this?

CONSTRUCTED RESPONSE:

Explain the step-by-step process of how a banana you eat for breakfast provides the energy for running in gym class. Include the roles of glucose, oxygen, and ATP.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every movement you make — every heartbeat, every thought, every slam dunk — is powered by ATP, a molecule produced through cellular respiration. Understanding this process reveals how food literally becomes action.

From Food to Glucose

When you eat, digestion breaks complex food molecules into simpler ones. Carbohydrates are broken down into glucose ($C_6H_{12}O_6$). This glucose enters the bloodstream and is delivered to every cell in your body. An average person's body contains about 4 grams of glucose in their blood at any time.

Cellular Respiration

The equation: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \sim 36$ ATP. This happens in the mitochondria of every cell. The process has three stages: glycolysis (splitting glucose), the Krebs cycle (extracting electrons), and the electron transport chain (making most of the ATP).

ATP: The Energy Currency

ATP is like cash for your cells. Every muscle contraction, nerve impulse, and chemical reaction is paid for with ATP. Your body produces and uses about 40 kg of ATP per day — roughly your body weight! But you only have about 250 grams at any given time, so it's constantly being recycled.

Aerobic vs. Anaerobic

With oxygen (aerobic): 1 glucose → ~36 ATP (efficient). Without enough oxygen (anaerobic): 1 glucose → 2 ATP + lactic acid (inefficient). During intense exercise, muscles outpace oxygen delivery and switch to anaerobic respiration. The lactic acid buildup causes the burning feeling in your muscles.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify food intake, oxygen supply, ATP production, and athletic output as system components.

E — Establish Relationships
Students determine that both food and oxygen are needed for ATP production (positive), and ATP drives athletic output (positive).

V — Visualize & Model
Students build the cellular respiration model showing how food and oxygen become energy for movement.

E — Evaluate Outcomes
Students compare resting, maximum effort, and no-fuel scenarios.

R — Revise & Extend
Students add lactic acid or heart rate to model anaerobic conditions.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"LeBron James eats a chicken sandwich. Two hours later, he dunks over three defenders. HOW?"*

Watch For

Play a LeBron highlight. Ask: Where did the energy for that come from?

Slide 2: Learning Objectives

3 min

Say: *"Today we trace the incredible journey from food to athletic performance."*

Watch For

Define cellular respiration as 'your cells burning fuel.'

Slide 3: Big Question

3 min

Say: *"A sandwich becomes a slam dunk. A banana becomes a sprint. How?"*

Watch For

Quick write: Trace the energy from lunch to running.

Slide 4: LEVER Framework

2 min

Say: *"We'll model how your body converts food into movement."*

Watch For

Preview the LEVER steps.

Slide 5: Activity 1: Components

8 min

Say: *"What inputs does your body need to produce energy?"*

Watch For

Sort components. Emphasize that you need BOTH food AND oxygen.

Slide 6: Activity 2: Connections

8 min

Say: *"Without oxygen, your body switches to plan B — but it's way less efficient."*

Watch For

Map the pathway from food to ATP to movement.

Slide 7: Activity 3: Simulation

10 min

Say: *"What happens when you sprint but haven't eaten? Or eat but don't have enough oxygen?"*

Watch For

Students run all three scenarios. Compare the graphs.

Slide 8: Discoveries

5 min

Say: *"Your body is a chemical factory that turns food into motion. Every. Single. Move."*

Watch For

Discuss: Why do you breathe harder during exercise?

Slide 9: STEM Challenge

12 min

Say: *"The basketball team hired you to design their game-day nutrition plan."*

Watch For

Teams design evidence-based nutrition plans.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Food Intake (Glucose), Oxygen Supply
- **INTERNAL:** ATP Production, Athletic Output

Relationships

- **Food Intake to ATP Production:** POSITIVE (+) — More glucose available means more fuel for cellular respiration to produce ATP.
- **Oxygen Supply to ATP Production:** POSITIVE (+) — More oxygen allows aerobic respiration, which produces ~18 times more ATP than anaerobic.
- **ATP Production to Athletic Output:** POSITIVE (+) — More ATP means more energy available for muscle contraction, speed, and power.

Key Discoveries

- Food is literally rearranged into energy through chemical reactions in your cells
- Cellular respiration: Glucose + Oxygen → CO2 + Water + ATP (energy)
- Without enough oxygen, cells switch to anaerobic respiration — less efficient, produces lactic acid
- Athletes need more food AND more oxygen to sustain high performance

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Food gives you energy directly"	Food doesn't power muscles directly. Digestion breaks food into glucose, which cells convert to ATP through respiration. ATP is the actual energy molecule that powers all cellular work.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"You only breathe to get oxygen"	You breathe for TWO reasons: to bring in O2 (input for respiration) AND to expel CO2 (waste product of respiration). Both are equally important. Too much CO2 is toxic.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Energy is created when you eat"	Energy is TRANSFORMED, not created. The chemical energy stored in food molecules (glucose) is converted to ATP energy through respiration. The total energy is conserved.